Notes: Malenko & Shen (RFS, 2016)

The Role of Proxy Advisory Firms: Evidence from a Regression-Discontinuity Design

Contents

1 Big picture

1.1 Question

Proxy advisors (especially ISS) are influential in shareholder voting, but there is disagreement about *why*:

- **Mechanical influence view:** investors "blindly follow" ISS, so ISS recommendations largely determine outcomes.
- **Information/intermediation view:** ISS aggregates information and lowers voting costs, so investors use it as a signal.

The paper quantifies **the causal effect** of ISS recommendations on voting outcomes and uses cross-sectional evidence to infer what that effect implies for ISS's informational role.

1.2 Setting

The analysis focuses on **say-on-pay (SoP)** votes for Russell 3000 firms in **2010–2011**:

- SoP became mandatory for most firms starting in 2011 under Dodd–Frank.
- These votes are advisory, but low support can trigger reputational costs, litigation risk, and heightened governance scrutiny.

1.3 Core identification idea

ISS follows an explicit **cutoff rule** for identifying underperforming firms based on total shareholder return (TSR). Firms just below vs. just above the cutoff are comparable in fundamentals, but are discontinuously more likely to receive a **negative ISS recommendation**. This yields a **fuzzy regression discontinuity design** (fuzzy RD) where the cutoff indicator instruments for receiving a negative recommendation.

1.4 Main conclusions

Falling just below the ISS TSR cutoff increases the probability of a negative ISS recommendation by about **15 percentage points** (first stage), and a negative recommendation reduces say-on-pay voting support by about **25 percentage points** (second stage). The effect is larger when the shareholder base is dominated by institutions that are more likely to rely on proxy advisors (e.g., more passive/less concentrated ownership), and smaller when ownership is more concentrated (blockholders/top institutions), consistent with heterogeneous reliance on ISS. Validity checks (density, covariate continuity, falsification samples) support the RD assumptions. Despite strong effects on votes, the paper finds little evidence of differential market reactions around the vote date or systematic changes in compensation terms post-vote in the local RD sample, suggesting that the primary detectable channel in this setting is via the voting outcome itself.

2 Data and construction of key variables

2.1 Sources

- ISS Voting Analytics: ISS recommendations and vote outcomes.
- Compustat/CRSP: TSR and firm characteristics.
- Thomson Reuters 13F: institutional ownership and concentration measures.
- GMI Ratings: governance controls (used in robustness/controls).

2.2 Outcome: voting support

Let $Votes_{it}$ be the percentage of votes cast "For" the say-on-pay proposal for firm i in year t:

$$Votes_{it} \in [0, 100] \quad \text{(percentage points).}$$

ISS recommendation indicator:

$$NegRec_{it} := \mathbf{1}\{\text{ISS recommends } \textit{Against}\}.$$

2.3 The ISS cutoff rule and the forcing variable

ISS underperformance criterion. ISS performs an in-depth qualitative review of compensation when a firm is classified as *underperforming*: both its one-year TSR and its three-year TSR are below the respective medians of its 4-digit GICS peer group.

Forcing variable and cutoff indicator. Define the cutoff indicator (Equation (1) in the paper):

$$BelowCutoff_{it} = \begin{cases} 1, & \text{if } MaxTSR_{it} < 0, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

The forcing variable is (Equation (2)):

$$MaxTSR_{it} = \max\Big(TSR_{it}^{(1)} - MedianTSR_{it}^{(1)},\, TSR_{it}^{(3)} - MedianTSR_{it}^{(3)}\Big), \tag{2}$$

where $TSR_{it}^{(n)}$ is the n-year TSR of firm i measured in year t, and $MedianTSR_{it}^{(n)}$ is the median n-year TSR in year t across all Russell 3000 firms in the same 4-digit GICS group.

Key interpretation. $MaxTSR_{it} < 0$ implies *both* one-year and three-year TSR are below the peer median (because the maximum of the two gaps is negative). Thus the cutoff at 0 is a clean implementation of the ISS "underperformer" rule.

3 Model block 1: fuzzy RD / 2SLS estimation of the ISS effect on votes

3.1 Why fuzzy RD?

Passing the TSR cutoff does not deterministically assign the ISS recommendation: ISS may still issue a positive recommendation below the cutoff after qualitative review, and may issue a negative recommendation above the cutoff for other reasons. Thus the cutoff creates a discontinuity in the *probability* of a negative recommendation, not a deterministic assignment.

3.2 Two-stage system (Equation (3))

The paper estimates the local average treatment effect (LATE) of a negative ISS recommendation on voting support using 2SLS:

$$\begin{aligned}
\textbf{First stage:}\quad NegRec_{it} &= \gamma_0 + \gamma_1 BelowCutoff_{it} + f_1(MaxTSR_{it}) \\
&\quad + BelowCutoff_{it} \cdot f_2(MaxTSR_{it}) + \gamma_X' X_{it} + u_{it}, \\
\textbf{Second stage:}\quad Votes_{it} &= \alpha_0 + \alpha_1 \widehat{NegRec}_{it} + g_1(MaxTSR_{it}) \\
&\quad + BelowCutoff_{it} \cdot g_2(MaxTSR_{it}) + \alpha_X' X_{it} + \varepsilon_{it}.
\end{aligned} \tag{3}$$

Here:

- $\widehat{NegRec}_{it}$ is the fitted value from the first stage,
- f_1, f_2, g_1, g_2 are smooth functions (local polynomials) of $MaxTSR$,
- X_{it} includes firm and compensation controls (Appendix Table A1 lists all definitions).

3.3 Identifying assumptions (local continuity)

The RD identification requires (locally around $MaxTSR = 0$):

(i) **Continuity of potential outcomes:** absent the discontinuity in ISS recommendation probability, the conditional expectation of outcomes would vary smoothly in $MaxTSR$.

(ii) **No manipulation of the forcing variable:** firms cannot precisely sort just above the cutoff; equivalently, the density of $MaxTSR$ is continuous at 0 (McCrary test).

(iii) **Exclusion restriction:** crossing the cutoff affects votes only through ISS recommendations (not through some other discontinuous channel).

(iv) **Monotonicity (for LATE):** falling below the cutoff weakly increases the likelihood of a negative recommendation for all firms (no "defiers").

4 Model block 2: informational role of ISS (stylized voting model)

To interpret why the OLS and RD estimates are similar, the paper presents a stylized model of an investor's vote.

4.1 Vote as a function of ISS and the investor's private signal (Equation (4))

Let $Signal^{SH}$ denote the shareholder's own signal about the value of the proposal (unobserved by the econometrician). The shareholder's voting decision can be written as:

$$Vote = \beta_0 + \beta_1 NegRec + \beta_2 Signal^{SH} + \varepsilon. \tag{4}$$

Here β_1 is the causal effect of ISS on voting (holding the shareholder's own signal fixed).

4.2 OLS omitted-variable bias (Equation (5))

Estimating $Vote$ on $NegRec$ by OLS omits $Signal^{SH}$, yielding an asymptotic bias:

$$Bias_{OLS} = \beta_2 \cdot Corr(NegRec, Signal^{SH}) \cdot \frac{Std.Dev.(Signal^{SH})}{Std.Dev.(NegRec)}. \tag{5}$$

4.3 Interpretation of the "small OLS–RD gap"

If the OLS and RD estimates are similar, then the bias term in (5) is small, which is consistent with:

- **Low independent information acquisition:** $Signal^{SH}$ is weak/absent (so β_2 or its variance is small), i.e., many shareholders mainly rely on ISS.
- **Different information sets:** shareholders have informative signals, but those signals are weakly correlated with ISS (small $Corr(NegRec, Signal^{SH})$), implying ISS and shareholders may be learning about different aspects of the proposal.

The paper uses heterogeneity evidence (Table 4) to distinguish these stories: the effect of ISS is smaller where independent research is more likely (concentrated ownership/blockholders) and larger where reliance on ISS is more plausible (more passive institutions).

5 Interpretation of all figures

5.1 Figure 1: discontinuities in ISS recommendations and vote support

Panel A plots the probability of a negative ISS recommendation against $MaxTSR$. There is a clear upward jump at 0, establishing a strong **first stage**. Panel B plots voting support against $MaxTSR$ and shows a downward jump at 0. **Graphical fuzzy-RD logic:** the LATE is approximately the ratio of the vote discontinuity to the recommendation discontinuity. The figure implies a large negative effect of a negative recommendation on voting support.

5.2 Figure 2: McCrary density test (no manipulation)

Figure 2 plots the histogram and estimated density of $MaxTSR$ around 0. The density is smooth at the cutoff and the McCrary statistic is not significant, supporting the assumption that firms do not precisely manipulate TSR to locate just above the cutoff.

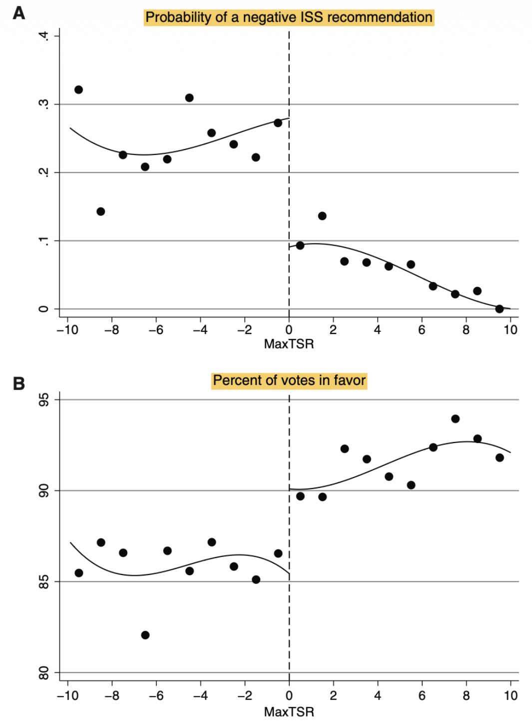

Figure 1
Probability of a negative ISS recommendation and voting support

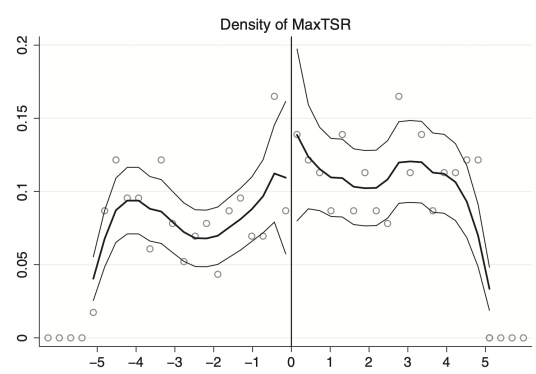

Figure 2
Density of the forcing variable

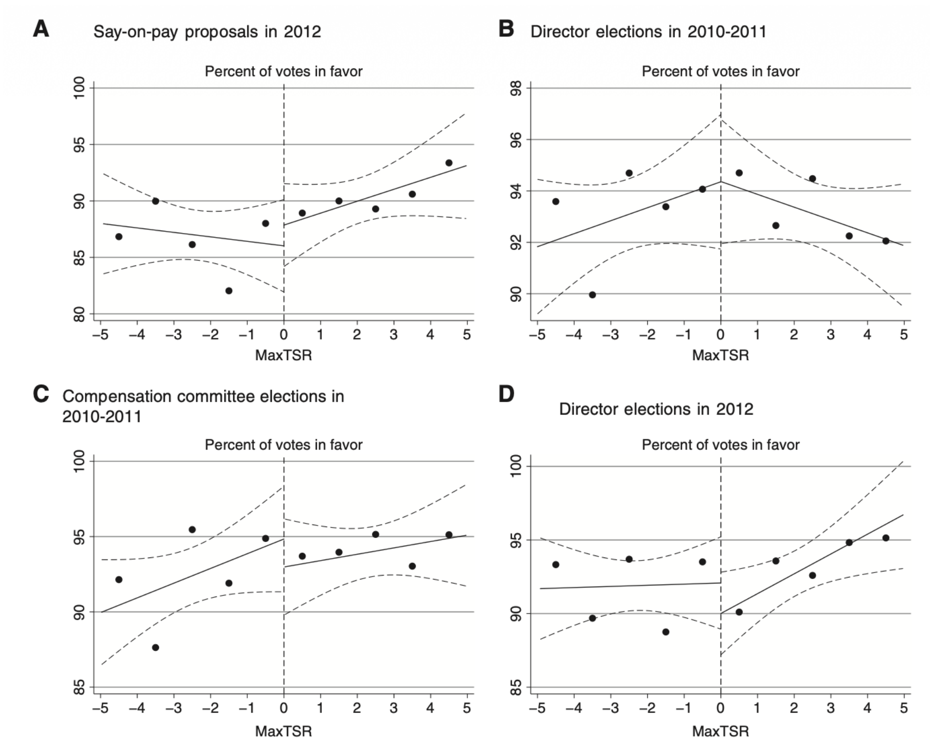

Figure 3
Falsification tests on alternative samples

5.3 Figure 3: falsification tests on samples where the cutoff rule should not apply

Figure 3 repeats the graphical analysis in settings where ISS's TSR cutoff rule is not relevant (e.g., years outside the 2010–2011 SoP sample or different election types). Voting support is continuous at the same cutoff in these placebo samples. **Interpretation:** the discontinuity is specific to the setting where the ISS cutoff rule is operative, strengthening causal interpretation.

6 Interpretation of all tables

6.1 Table 1: descriptive statistics

- Panel A compares firm and compensation characteristics in the full sample versus the 5% RD bandwidth sample, and reports p-values for mean differences. The key message is that the local sample looks broadly comparable and economically reasonable (no dramatic reweighting).
- Panel B shows the distribution of voting support by ISS recommendation. Votes are much lower when ISS recommends "Against," and nearly all proposals pass overall (SoP votes rarely fail).

6.2 Table 2: first stage (probability of a negative ISS recommendation)

Table 2 estimates the first stage in (3) using a linear probability model:

Table 1
Descriptive statistics

A

	Full sample					5% bandwidth					Diff. in means
	Mean	25th	50th	75th	SD	Mean	25th	50th	75th	SD	p-val
Market value of equity	5.53	0.44	1.15	3.50	17.78	7.97	0.64	1.64	4.37	26.47	0.08
ROA	0.02	0.00	0.03	0.07	0.15	0.03	0.01	0.03	0.07	0.12	0.19
Leverage	0.20	0.04	0.16	0.31	0.20	0.22	0.06	0.18	0.32	0.19	0.15
Market-to-book	1.90	1.04	1.35	2.06	1.68	1.59	1.02	1.26	1.78	0.92	0.00
Institutional ownership	0.72	0.58	0.76	0.89	0.23	0.71	0.57	0.75	0.88	0.22	0.65
Institutional ownership HHI	0.06	0.04	0.05	0.07	0.06	0.06	0.04	0.05	0.06	0.07	0.62
CEO total compensation	4.76	1.46	3.00	6.20	4.96	5.44	1.73	3.64	6.93	5.17	0.01
Proportion of stock-based comp.	0.40	0.23	0.42	0.57	0.27	0.42	0.27	0.45	0.57	0.23	0.29
Growth in CEO total comp.	0.47	−0.07	0.18	0.59	1.62	0.42	−0.10	0.12	0.48	1.55	0.58

B

	Voting support					Number of observations by voting outcome		
ISS recommendation	Mean	10th	50th	90th	SD	Fail	Pass	Total
Against	68.9%	48.5%	68.9%	88.7%	15.2%	29	227	256
For	93.2%	83.3%	95.6%	99.1%	7.0%	0	1,764	1,764
Total (against and for)	90.1%	73.5%	94.9%	99.0%	11.7%	29	1,991	2,020

Table 2
Probability of a negative ISS recommendation (first stage)

	(1)	(2)	(3)	(4)	(5)
	NegRec	NegRec	NegRec	NegRec	NegRec
BelowCutoff	0.145**	0.146**	0.146**	0.148**	0.142**
	(0.068)	(0.069)	(0.070)	(0.067)	(0.067)
MaxTSR	−0.008	−0.010	−0.017	−0.012	−0.014
	(0.012)	(0.016)	(0.016)	(0.016)	(0.016)
BelowCutoff · MaxTSR		0.004	0.013	0.009	0.010
		(0.024)	(0.024)	(0.023)	(0.023)
CEO total compensation				0.007*	0.018***
				(0.004)	(0.006)
Growth in CEO total compensation				0.062***	0.061***
				(0.012)	(0.012)
Proportion of stock-based compensation				−0.064	−0.060
				(0.093)	(0.095)
Institutional ownership				−0.007	0.004
				(0.096)	(0.096)
Insider ownership				0.412***	0.360***
				(0.125)	(0.127)
Log(market value of equity)					−0.043**
					(0.020)
ROA					−0.133
					(0.172)
M/B					0.024
					(0.023)
Year FE	No	No	Yes	Yes	Yes
Industry FE	No	No	Yes	Yes	Yes
Observations	403	403	403	394	393
R^2	0.062	0.062	0.076	0.184	0.198
F-statistic	4.49	4.52	4.33	4.85	4.45

Table 3
Effect of ISS recommendations on voting outcomes (second stage)

	(1)	(2)	(3)	(4)	(5)
	Votes	Votes	Votes	Votes	Votes
NegRec	−24.620**	−25.111**	−29.001**	−28.717***	−26.555**
	(11.210)	(11.160)	(11.588)	(10.655)	(10.853)
MaxTSR	0.045	0.175	0.129	0.125	0.226
	(0.358)	(0.450)	(0.531)	(0.455)	(0.463)
BelowCutoff·MaxTSR		−0.294	−0.302	−0.198	−0.277
		(0.560)	(0.585)	(0.544)	(0.534)
CEO total compensation				−0.273**	−0.677***
				(0.120)	(0.241)
Growth in CEO total compensation				0.261	0.167
				(0.720)	(0.717)
Proportion of stock-based compensation				−0.047	−0.210
				(2.228)	(2.218)
Institutional ownership				−5.597**	−6.009***
				(2.246)	(2.186)
Insider ownership				13.023**	14.370***
				(5.292)	(4.861)
Log(market value of equity)					1.658**
					(0.665)
ROA					5.750
					(4.247)
M/B					0.353
					(0.585)
Year FE	No	No	Yes	Yes	Yes
Industry FE	No	No	Yes	Yes	Yes
Observations	403	403	403	394	393
R^2	0.547	0.547	0.536	0.605	0.628
OLS coefficient on NegRec	−25.339***	−25.338***	−25.052***	−26.070***	−25.443***
	(1.185)	(1.186)	(1.227)	(1.234)	(1.222)
Durbin-Wu-Hausman test *p*-value	0.949	0.984	0.720	0.795	0.915

- The coefficient on $BelowCutoff$ is about **0.14–0.15**, meaning falling below the cutoff increases the probability of a negative recommendation by roughly **15 percentage points**.
- Specifications add smooth controls in $MaxTSR$, year and industry fixed effects, and additional controls; the discontinuity remains stable.
- The reported first-stage F-statistics are modest (around 4–5 in the baseline 5% bandwidth), motivating the instrument-strength diagnostics in Table 7.

6.3 Table 3: second stage (effect of ISS on voting outcomes)

Table 3 estimates the second stage in (3):

- The coefficient on $NegRec$ is around **−25 to −29** percentage points: a negative ISS recommendation reduces SoP support by about **25 pp**.
- The table also reports OLS coefficients on $NegRec$ (around –25) and Durbin–Wu–Hausman p-values, indicating little evidence of endogeneity of $NegRec$ within the local RD sample.
- Interpretation: in the close-to-cutoff region, ISS recommendations have a large, causal effect on the aggregate vote share.

6.4 Table 4: heterogeneity — when is ISS more influential?

Table 4 splits the sample by investor-base characteristics and re-estimates the fuzzy-RD effect:

Table 4
Variation in the effect of ISS across firms

	(1)		(2)		(3)		(4)		(5)	
	Inst.ownership HHI		% Inst.blockholders		% Top 10 institutions		% Passive institutions		Inst.ownership	
	Low	High	Low	High	Low	High	Low	High	Low	High
BelowCutoff	0.164**	0.158**	0.126*	0.169**	0.163**	0.156**	0.201***	0.159**	0.176**	0.187***
(First stage)	(0.071)	(0.066)	(0.070)	(0.073)	(0.070)	(0.066)	(0.070)	(0.066)	(0.068)	(0.069)
NegRec	−41.12***	−21.27*	−35.72**	−25.56**	−37.48***	−23.38**	−18.87**	−38.20***	−24.08***	−32.30***
(Second stage)	(12.07)	(11.04)	(16.33)	(11.95)	(11.20)	(11.21)	(7.78)	(11.76)	(8.65)	(10.19)
MaxTSR controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	386	386	337	336	385	387	384	384	388	389

Table 5
Robustness

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
	3%	4%	5%	6%	7%	8%	9%	10%	20% quadratic	20% cubic	20% quartic
BelowCutoff	0.18**	0.17**	0.15**	0.19***	0.19***	0.19***	0.20***	0.18***	0.19***	0.19***	0.21***
	(0.09)	(0.08)	(0.07)	(0.06)	(0.06)	(0.05)	(0.05)	(0.05)	(0.05)	(0.07)	(0.08)
NegRec	−20.33*	−24.68**	−29.00**	−27.99***	−21.09***	−23.47***	−23.29***	−28.56***	−22.38***	−23.62***	−23.91**
	(12.01)	(10.26)	(11.59)	(8.33)	(7.81)	(7.08)	(6.37)	(6.93)	(7.07)	(9.16)	(10.18)
Observations	238	313	403	490	574	651	717	785	1,244	1,244	1,244

- **Ownership concentration (HHI, blockholders, top-10 institutions):** the effect of $NegRec$ is *smaller in magnitude* when ownership is more concentrated or dominated by large blockholders/top institutions (consistent with these investors being more likely to do independent analysis).
- **Passive institutions:** the effect is *larger* when the share held by passive/"transient" investors is higher (consistent with greater reliance on ISS).
- **Total institutional ownership:** the effect is larger in the high-institutional-ownership subsample, consistent with institutions being the marginal voters and with ISS's role in institutional voting.

Overall, Table 4 supports the view that ISS influence is strongest when shareholders are most likely to delegate information acquisition to proxy advisors.

6.5 Table 5: robustness to bandwidths and functional forms

Table 5 varies:

- RD bandwidth (3% to 20% around the cutoff), and
- polynomial order (quadratic/cubic/quartic for a fixed 20% bandwidth).

Main message:

- The first-stage discontinuity remains sizable across bandwidths.
- The second-stage effect of $NegRec$ remains negative and economically large, with magnitudes that are broadly stable.

6.6 Table 6: covariate continuity around the cutoff (balance tests)

Table 6 compares means of key firm and compensation variables just below versus just above the cutoff and reports RD coefficients for these covariates. The lack of systematic discontinuities supports the core RD assumption that treated and control firms are locally comparable. This directly addresses concerns that the cutoff might proxy for other sharp changes in fundamentals.

Table 6
Distribution of firm characteristics around the cutoff

	RD coeff. on BelowCutoff	$(-5\%, 0)$ Mean	$(-5\%, 0)$ Obs	$(0, 5\%)$ Mean	$(0, 5\%)$ Obs	Diff. in means p-val
Log(market value of equity)	−0.02 (0.29)	0.71	173	0.58	230	0.39
M/B	−0.10 (0.15)	1.56	173	1.62	229	0.49
ROA	−0.03 (0.02)	0.03	173	0.03	230	0.82
Leverage	0.00 (0.04)	0.21	165	0.22	216	0.74
CEO total compensation	0.06 (0.96)	5.81	173	5.17	230	0.22
CEO compensation percentile	1.29 (5.21)	60.3	173	57.9	230	0.38
Proportion of stock-based comp.	0.06 (0.04)	0.42	173	0.42	230	0.91
Proportion of incentive pay	0.02 (0.05)	0.53	170	0.55	226	0.51
1-year growth in CEO total comp.	0.06 (0.29)	0.44	172	0.41	227	0.85
3-year growth in CEO total comp.	−0.03 (0.04)	0.05	154	0.05	207	0.91
Institutional ownership	−0.03 (0.04)	0.72	173	0.71	230	0.62
Insider ownership	0.01 (0.03)	0.12	169	0.09	226	0.12

Table 7
Instrument strength

	5%	6%	7%	8%	9%	10%
First-stage F-statistic	4.33	8.55	10.46	12.71	16.06	14.32
Anderson-Rubin p-value	0.086	0.018	0.055	0.018	0.009	0.003
Reduced-form regression	−4.22*	−5.20**	−3.91**	−4.45**	−4.63***	−5.07***
	(2.42)	(2.17)	(1.99)	(1.84)	(1.74)	(1.65)
Observations	403	490	574	651	717	785

6.7 Table 7: instrument strength diagnostics

Given relatively modest first-stage F-statistics in narrow bandwidths, Table 7 reports:

- first-stage F-statistics for wider bandwidths (5%–10%), which rise substantially (reducing weak-instrument concerns),
- Anderson–Rubin p-values for the second-stage coefficient (robust inference under weak instruments),
- reduced-form estimates of the cutoff effect on votes.

Interpretation: while the 5% bandwidth is the sharpest local design, broader windows deliver stronger instruments with similar first-stage magnitudes, and inference remains consistent with a large negative effect of negative ISS recommendations.

7 Additional results (Section 5.2): ex post outcomes

The paper examines whether falling below the cutoff affects:

- stock-price reactions around the vote date, and
- subsequent compensation practices (e.g., CEO pay changes, pay mix, severance/benefits).

In the local RD sample, these ex post outcomes do not exhibit robust discontinuities. Two interpretations are consistent with this finding:

(i) markets may largely anticipate the vote and/or interpret SoP votes as conveying limited incremental information in this narrow region;

(ii) firms may respond to low support through channels not captured by the specific compensation variables examined over the measured horizon (or effects may be too small to detect with RD power).

8 Bottom-line takeaway

The paper isolates a clean, policy-rule-based source of variation in ISS recommendations using a fuzzy RD design. A firm that barely falls below ISS's TSR cutoff becomes about 15pp more likely to receive a negative recommendation, and a negative recommendation causally reduces say-on-pay support by about 25pp. The magnitude varies predictably with the shareholder base: ISS matters less when large informed blockholders dominate and more when ownership is diffuse and passive institutions are important. The RD validity checks (density, covariates, falsification) support the design, making the estimates a credible causal benchmark for ISS influence in SoP voting.

9 Conclusion and intuition

9.1 Coherent summary: what the paper concludes

This paper provides causal evidence that proxy advisory firms—in particular ISS—have a *large and economically meaningful* impact on shareholder voting outcomes. The core empirical challenge is that ISS recommendations are endogenous: they are correlated with firm fundamentals and with shareholders' own assessments of compensation quality, so simple correlations cannot be interpreted causally.

The authors overcome this problem by exploiting a specific feature of ISS's (historical) say-on-pay (SoP) voting guidelines: ISS first classifies firms as "underperformers" using a *cutoff rule* based on one-year and three-year total shareholder returns (TSR) relative to industry peers. Firms just below versus just above this cutoff are very similar in fundamentals but are discontinuously more likely to be flagged for deeper review and, consequently, to receive a negative ISS recommendation. This generates a *fuzzy regression discontinuity design* that instruments for receiving a negative ISS recommendation.

The main conclusions are:

(i) **ISS recommendations causally move votes by a large amount.** In the local RD sample, a negative ISS recommendation reduces the percentage of votes cast in favor of the SoP proposal by about **25 percentage points** relative to a positive recommendation.

(ii) **ISS influence is heterogeneous and aligns with "delegated voting" patterns.** ISS has a stronger effect in firms with:

- higher institutional ownership,
- more *dispersed* institutional ownership (less concentrated),
- a larger fraction of shares held by institutions with *small stakes* or *high turnover*.

The influence is weaker when ownership is more concentrated (e.g., large blockholders/top institutions), consistent with greater independent analysis by large, engaged investors.

(iii) **Identification validity is supported by standard RD diagnostics.** The distribution of the forcing variable is smooth at the cutoff (no evidence of sorting/manipulation), and covariates are locally balanced. Placebo/falsification exercises show that discontinuities appear only where the ISS cutoff rule is relevant.

(iv) **Implication for the policy debate: ISS influence is real, not purely spurious correlation.** The results imply that the high correlation between ISS recommendations and voting outcomes is not only due to ISS and shareholders independently reaching the same conclusions using the same information. Instead, in settings where many shareholders face information and processing costs, ISS recommendations can act as a pivotal input into voting.

9.2 Economic intuition: why the cutoff-based RD is powerful and what it reveals about ISS

1) Why the cutoff generates quasi-random variation. Around the cutoff, firms differ only infinitesimally in measured TSR performance. If firms cannot precisely manipulate the forcing variable, then being just below versus just above the cutoff is approximately random. However, ISS's guideline rule creates a discrete jump in the probability of a negative recommendation at that point. Thus, the cutoff provides an exogenous shift in recommendations that can be used to trace a causal effect on voting outcomes.

2) Why the design is fuzzy rather than sharp. Crossing the cutoff does not deterministically produce a negative recommendation, because ISS may still issue a positive recommendation after qualitative review, and may also issue negative recommendations above the cutoff for other concerns. Therefore, the discontinuity changes the *likelihood* of a negative recommendation rather than assigning it perfectly. The causal effect is identified via a fuzzy RD / 2SLS logic: the vote-share discontinuity at the cutoff is scaled by the recommendation discontinuity.

3) What the heterogeneity implies about the role of proxy advisors. The strongest effects appear where shareholders are more likely to rely on standardized advice: diffuse institutional ownership, many small-stake investors, and high-turnover institutions. When ownership is concentrated, large shareholders have stronger incentives to collect information themselves, making them less sensitive to ISS. This pattern supports an *intermediation / delegated-information* interpretation: ISS reduces voting frictions and shapes outcomes especially when the marginal voter is information-constrained.

4) Why "ISS matters" is not the same as "ISS is always correct." The paper identifies ISS's causal influence on votes, not the normative optimality of ISS guidance. In economic terms, the results show that ISS is a central node in the voting process and can shift equilibrium voting outcomes, which is crucial for understanding corporate governance in practice and for evaluating regulatory proposals that target proxy advisors.

9.3 Final takeaway

Using a fuzzy regression discontinuity from an ISS cutoff rule, Malenko and Shen show that a negative ISS recommendation causally lowers say-on-pay support by roughly 25 percentage points, with larger effects when ownership is more institutional and more dispersed, consistent with shareholders delegating information acquisition to proxy advisors.